



BTS-IV-04 19-0647

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B.Tech. Degree IV Semester Examination April 2019

CE 15-1402 SURVEYING II (2015 Scheme)

Time : 3 Hours

Maximum Marks : 60

PART A (Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) What are the different elements of a compound curve?
(b) How do you arrive at the length of a transition curve?
(c) Discuss the classification of triangulation system.
(d) Write short note on elevated tower.
(e) State the theory of least squares.
(f) What is celestial sphere?
(g) Define latitude and longitude of a place.
(h) What are the uses of hydrographic survey?
(i) What are the advantages of echo – sounding?
(j) What is distortion of the vertical photograph?

PART B

(4 × 10 = 40)

- II. (a) Write the principle and procedure of Rankine's method for setting out a circular curve. (6)
(b) Explain different types of vertical curves with figure. (4)
- OR**
- III. (a) A compound curve is to connect two straights having a deflection angle of 90°. As determined from the plan, the lengths of the two tangents are 350 meters and 400 meters respectively. Calculate the lengths of the two arcs if the radius of the first curve is to be 300 meters. (7)
(b) What is cant deficiency? (3)
- IV. (a) The altitudes of two proposed stations A and B, 90 km apart are respectively 225 m and 550 m. The intervening obstructions situated at C, 40 km from A has an elevation of 287 m. If A and B are intervisible, and if necessary, find by how much B should be raised so that the line of sight must nowhere be less than 3 m above the surface of the ground. (6)
(b) State the laws of weight. (4)

OR

(P.T.O.)

- V. (a) Given the following equations (7)

$$\begin{array}{lll} A & = 42^{\circ}36'28'' & \text{wt } 2 \\ B & = 28^{\circ}12'42'' & \text{wt } 2 \\ C & = 65^{\circ}25'16'' & \text{wt } 3 \\ 2A+3B & = 99^{\circ}01'59'' & \text{wt } 2 \\ 2B+C & = 159^{\circ}03'11'' & \text{wt } 1 \end{array}$$

Find the most probable values of A, B and C.

- (b) Explain the figure adjustment of a plane triangle. (3)

- VI. (a) The local sidereal time at a place (Longitude $112^{\circ}20'15''\text{W}$) is $18^{\text{h}} 28^{\text{m}} 12^{\text{s}}$. Calculate the corresponding L.M.T., given that G.S.T at G.M.M. is $8^{\text{h}} 10^{\text{m}} 28^{\text{s}}$ on that day. (6)

- (b) Explain solar time and sidereal time. (4)

OR

- VII. (a) Determine the azimuth of the sun during sunrise at a place having latitude $40^{\circ}30'\text{S}$. The declination of the sun is $25^{\circ}15'\text{N}$. (6)

- (b) Explain equation of time. (4)

- VIII. (a) Explain any four methods of sounding. (5)

- (b) What is the principle of terrestrial photogrammetry? (5)

OR

- IX. (a) Vertical photographs are taken from a height of 3000 m, for terrain lying at elevation 180 m, the focal length of the camera lens being 15 cm. The prints are 22.86×22.86 cm and the longitudinal overlap 60%, and the side lap is 30%. Determine the number of photographs required to cover an area of $8 \text{ km} \times 12.5 \text{ km}$. (5)

- (b) Explain the principle of binocular vision and stereoscopic fusion. (5)
